



(Year 2 / A2 Level)

Quantum Physics Basics Lecture Notes

edited by Learnmates

originally made by *Yuhao Yang*

CHAPTER 22

Quantum Physics Basics

In this chapter, we will study the laws of nature at very small scales.

We start by reviewing classical concepts like particles or waves, and then see how they break down when being applied to the microscopic world. For those observations that could not be explained with classical physics, we need *quantum physics*. The theory of quantum physics is very deep and profound, so surely way beyond the scope of our course. What we will study in this A-Level course are merely four tips of an enormous iceberg, namely

- photon theory / photoelectric effect
- matter waves / electron diffraction
- electron energy levels / atomic spectra
- band theory / electrical conductivity of solids

22.1 classical theories

in classical physics, both particle models and the wave models have been very useful

though both being successful, the two model are distinct in many aspects

to understand a phenomenon, we take either the particle picture or the wave picture

it seems that there is no way particle models would reconcile with wave models, or is it not?

22.1.1 particle models

in particle model, any system is considered to consist of particles governed by *Newtonian mechanics*, physical properties of this system are predicted by studying behaviour of particles

➤ areas of science where particle models are used to interpret and make predictions include:

- *electricity*: electric current formed by motion of charge carriers

- *ideal gas*: pressure caused by collisions of gas molecules
- *solid*: elasticity due to interaction between solid atoms
- *radioactivity*: decay interpreted as the emission of α -/ β -particles from the nucleus
- *chemistry*: chemical reactions due to exchange of electrons between atoms and molecules
- ...

22.1.2 wave models

in the wave picture, energy is transferred via the vibration of medium or force fields

➤ phenomena that can be explained in terms of wave models include:

- *water waves*: variation in the vertical displacement of water surface
- *propagation of sound*: variation in the pressure and density of medium
- *light*: variation of electric and magnetic fields
- ...

➤ a key feature that makes waves distinct from particles is that waves can *superpose*
when two or more waves meet, they add up or cancel out to give a resultant wave
this gives rise to the characteristic properties of waves:

- *interference*: waves superpose to form a resultant wave of greater or lower amplitude
- *diffraction*: bending of waves around obstacles, or spreading out of waves through slits

22.1.3 history of light

efforts to understand the nature of light can be traced all the way back to ancient Greeks

we are not going to examine the ideas of early thinkers well over 3,000 years ago

let's skip a couple of years and jump to the ideas developed since the Scientific Revolution

➤ particle theory of light (*Issac Newton*, 1671)

key idea: light rays is comprised of a stream of massless particles called *corpuscles*

- explains straight-line propagation
- explains reflection and refraction
- explains colours of light seen in dispersion in prism (corpuscles have different colours)

the problems with the particle model include:

- does not agree with observations on refraction
- cannot predict the interference and diffraction of light

➤ wave theory of light (*Christian Huygens*, 1678)

key idea: light is a wave that transfers energy within a medium known as *aether*

- follows laws of reflection and refraction
- explains colours of light (light have different wavelengths)
- explains interference (Thomas Young's double slit experiment)
- explains diffraction (Poisson spot experiment)

only problem is that aether, the medium in which light lives, was not experimentally found

➤ electromagnetic theory (*James Maxwell*, 1865)

key idea: light is an *electromagnetic wave* ^[1]

- electric and magnetic fields travel through space in the form of waves at speed of light
- propagation of electromagnetic wave does not require medium, hence no need for aether

all behaviour of light known at that time could be explained with Maxwell's theory

so scientists were convinced that light travelled through space as an electromagnetic wave

22.2 photon theory

22.2.1 photoelectric effect

electromagnetic radiation incident upon a metal can cause emission of electrons from the metal surface, this is called **photoelectric effect**, ^[2] emitted electrons are called **photoelectrons**

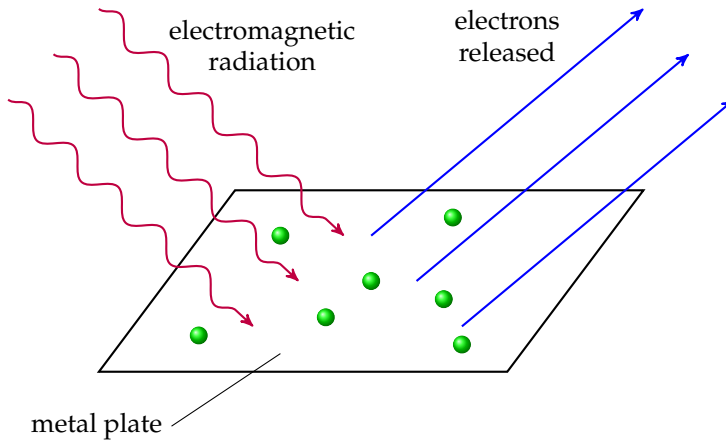
simply put, conduction electrons in metal gain additional energy by absorbing the incoming

^[1] *Maxwell's equations* fully describe the behaviour of electric and magnetic fields. Based on the four equations that Maxwell established, he found the vibration of electric and magnetic field can propagate in space as a classical wave. The speed of electromagnetic wave is given by

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = \frac{1}{\sqrt{8.85 \times 10^{-12} \times 4\pi \times 10^{-7}}} = 3.00 \times 10^8 \text{ m s}^{-1},$$

which is the same as speed of light. This gave Maxwell the intuition that light is an electromagnetic wave.

^[2] Photoelectric effect was first observed in 1887 by *Heinrich Hertz*, who found electrodes illuminated with ultraviolet light create electric sparks. In school labs, one can shine ultraviolet radiation from a mercury lamp onto a zinc plate to cause photo-emission.



radiation, if this energy is sufficient for electrons to overcome the electrostatic attraction from the positive metal ions, they break free from the metal surface

➤ some detailed experimental observations on the photoelectric effect are:

- there exists a minimum **threshold frequency** f_0 for incident radiation
when $f < f_0$, no electrons released from metal surface
- emission of electrons is immediate when radiation is incident as long as $f > f_0$
even low-intensity light is effective
- increasing intensity has no effect on energies of electrons
- increasing intensity of incident light causes number of photoelectrons emitted to increase
- increasing radiation frequency increases electron energies

➤ wave theory of light fails to explain any of these properties, according to wave model:

- electrons could gradually build up energies by absorbing wave energy over time
so radiation at any frequency should all work
- need very intense light to have immediate effect
- greater intensity mean higher energy, electrons released should have greater K.E.
- varying frequency of radiation should have no effect on energy of electrons released

photoelectric effect sees the breakdown of the wave model, some new ideas are needed!

22.2.2 photon theory

photoelectric effect was explained by *Albert Einstein* in 1905 [3]

Einstein's revolutionary idea: when radiation delivers energy to matter, the transfer of energy is not continuous but carried in *discrete* packets, called *photons*

a **photon** is a packet (*quanta*) of electromagnetic energy

➤ energy of one photon is given by: $E = hf$, where $h = 6.63 \times 10^{-34}$ J s is the **Planck constant**

since $E \propto f$, higher/lower radiation frequency means greater/smaller photon energies

➤ wave equation $c = \lambda f$ relates frequency f of an electromagnetic wave to its wavelength λ

so energy of one photon is also given by: $E = \frac{hc}{\lambda}$

since $E \propto \frac{1}{\lambda}$, longer/shorter wavelength means lower/greater photon energies

➤ the equation $E = hf$, or $E = \frac{hc}{\lambda}$, relates a particle property with a wave property

E is the energy of one photon, treated like a single particle

but f and λ are both introduced to describe a wave, not a particle

➤ intensity of a beam of radiation is: $I = \frac{P}{A}$

total energy incident on a given area per unit time determines the radiation intensity

so intensity depends on the product of the number of photons arriving per unit time and

energy of each photon: $I \propto nhf$, or more precisely, $I = \frac{nhf}{A}$

➤ when considering energy of a photon, **electronvolt** is a useful energy unit

one electronvolt (1 eV) is work needed to make an electron travel through a p.d. of 1 V

conversion between electronvolt and joule is: $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$

Example 22.1 A laser emits red light of 650 nm at a power rating of 2.0 mW. (a) What is the energy carried by one photon? (b) How many photons are emitted per second?

✎ energy of one photon: $E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{650 \times 10^{-9}} \approx 3.06 \times 10^{-19} \text{ J}$

number of photons: $N = \frac{\text{total energy output from laser}}{\text{energy of one photon}} = \frac{2.0 \times 10^{-3}}{3.06 \times 10^{-19}} \approx 6.54 \times 10^{15}$ □

[3] Albert Einstein was awarded the Nobel Prize in physics in 1921 for 'his discovery of the law of the photoelectric effect'. This discovery led to the quantum revolution of modern physics.

Question 22.1 Calculate the energy, in eV, of a photon of light of wavelength 440 nm.

Question 22.2 If the light source in Example 22.1 is replaced by a green laser with the same power output, how does the number of emitted photons per unit time change?

22.2.3 photoelectric effect explained

from the viewpoint of photon theory, photoelectric effect can be explained easily
to release an electron from metal, it requires a minimum energy Φ , called the **work function**,
for the electron to overcome attraction due to metal ions to escape from metal surface
as radiation shines upon metal, photon energies are absorbed by electrons
since photon energies are *discrete*, or *quantised*, the absorption is sort of all or nothing
if photon energy is greater than Φ , electrons break free from metal
excess energy, if any, would become kinetic energy of the free electron
this is summarised in **Einstein's photoelectric equation**: $hf = \Phi + E_{k,\max}$

➤ note that we are talking about *maximum* K.E. of emitted electrons

electron emitted from the *surface* would have greatest K.E.

for electrons to be released from *below* the surface, they require more energy than work function, so less K.E. than maximum value

➤ at critical condition, incoming photon has just enough energy to release electron

so at threshold frequency f_0 , photon energy equals work function: $hf_0 = \Phi$

➤ experimental observations mentioned in §22.2.1 can now be understood

- below threshold frequency f_0 , not enough photon energy available to electron to overcome work function, so no effect for $f < f_0$
- interaction between photon and electron is *one-to-one*, so no time delay
- greater radiation intensity means more photons per unit time, so more electrons released
- greater frequency means higher photon energy, so greater K.E. for emitted electron

Example 22.2 Given that work function energy of gold is 4.9 eV. Find the longest wavelength of electromagnetic wave that could release electrons from gold.

✍ threshold frequency: $f_0 = \frac{\Phi}{h} = \frac{4.9 \times 1.60 \times 10^{-19}}{6.63 \times 10^{-34}} \approx 1.18 \times 10^{18} \text{ Hz}$

threshold wavelength: $\lambda_0 = \frac{c}{f_0} = \frac{3.00 \times 10^8}{1.18 \times 10^{15}} \approx 2.54 \times 10^{-7} \text{ m}$ (ultraviolet light) □

Question 22.3 Sodium has a work function of $3.8 \times 10^{-19} \text{ J}$. If a light of 500 nm is incident on sodium, determine whether electrons can be emitted from the surface.

Question 22.4 When electromagnetic radiation of wavelength 1200 nm is incident on a metal surface, the maximum kinetic energy of the electrons released is found to be $5.4 \times 10^{-20} \text{ J}$. What is the work function of this metal?

Question 22.5 When a beam of light of a particular frequency and intensity is shone onto a metal surface, electrons are released. If another beam of light of same intensity but higher frequency is used, what is the effect on the rate of emission of electrons from this surface?

measurement of the Planck constant and work function energy

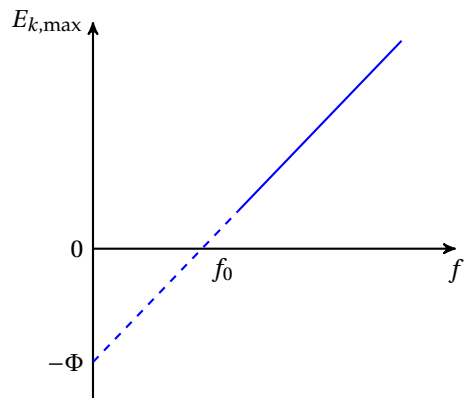
when radiation with different frequencies is incident onto a metal, we measure maximum K.E. of the electrons emitted from the surface, a set of readings $(f, E_{k,\max})$ can be found

note that the photoelectric equation can be rearranged as: $E_{k,\max} = hf - \Phi$

if we plot a graph of $E_{k,\max}$ against f , data points shall fall on a straight line

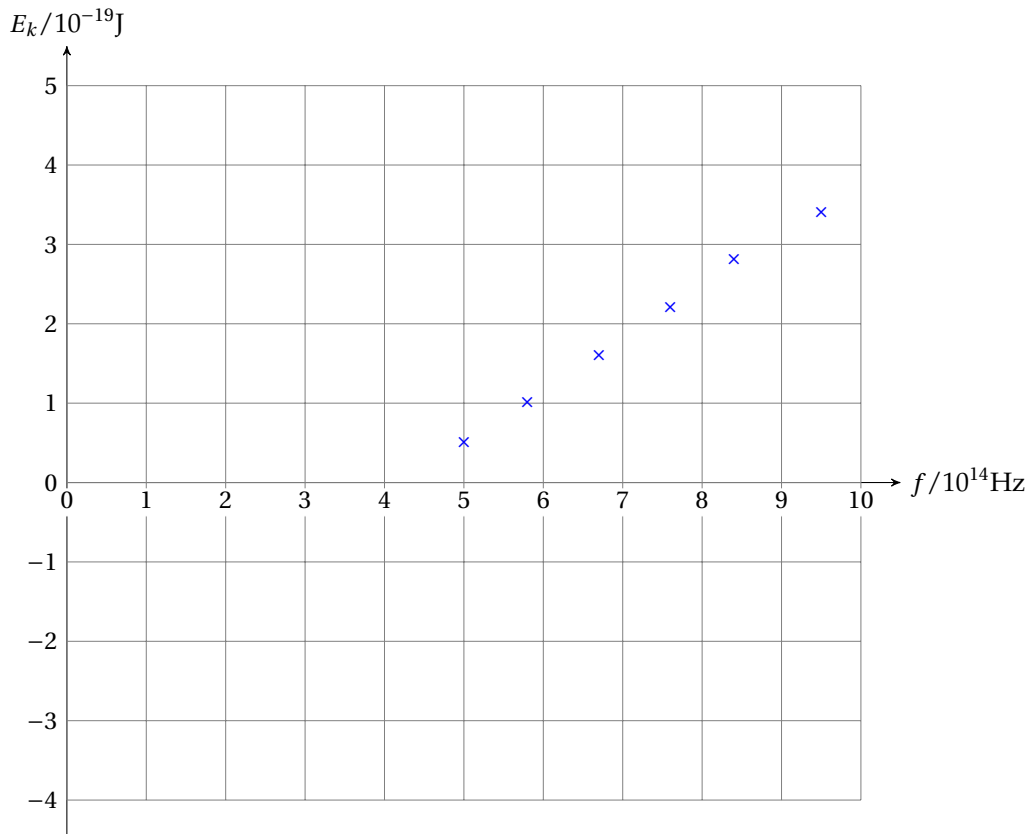
information about the Planck constant h , threshold frequency f_0 , work function Φ can all be computed with the best-fit line

- gradient = h
- y-intercept = $-\Phi$
- x-intercept = $\frac{\Phi}{h} = f_0$



Question 22.6 If a different metal with a greater work function energy is used, describe the change for the line that shows the variation with f of $E_{k,\max}$ for this metal.

Question 22.7 Electromagnetic radiation is incident upon a metal plate. The graph shows how maximum kinetic energy E_k of emitted electrons varies with frequency f of the radiation. Use the graph to find (a) the threshold frequency, (b) a value of Planck constant.



22.2.4 photon momentum

in photoelectric effect, photons can knock electrons out of a metal, this suggests that photons could have *momentum*, even though they do not have mass

earliest experimental evidence of photon momentum was came from *Arthur Compton* in 1923, who studied the scattering of X-ray photons by electrons in substances^[4]

it can be shown^[5] that photon momentum is given by: $p = \frac{h}{\lambda}$

➤ photon momentum a *relativistic* momentum, as photons move at speed of light

definition for classical momentum $p = mv$ does not apply for photons

^[4]Arthur Compton was awarded the Nobel Physics Prize in 1929 for the discovery of this scattering effect, now known as *Compton scattering*.

^[5]This derives from Einstein's theory of *special relativity*, which states that energy and momentum are related by the equation: $E^2 = m_0^2 c^4 + p^2 c^2$. Photons have zero rest mass, i.e., $m_0 = 0$. This relativistic relation becomes $E = pc$ for photons. Now recall photon energy is given by $E = hf$. Rearrange the terms, we can show: $p = \frac{hf}{c} = \frac{h}{\lambda}$.

➤ due to exchange of momentum, electromagnetic wave can exert *radiation pressure*

forces generated by radiation pressure are negligible under everyday circumstances, but they could have noticeable effects on spacecraft in outer space and comet tails

Example 22.3 A beam of light has wavelength 600 nm, cross-sectional area 0.16 cm^2 and power 5.0 mW. The beam is normally incident onto a surface and is completely absorbed. Calculate, for a time of 1.0 s, (a) the number of photons incident onto the surface, (b) the change of total momentum of the photons, (c) the light pressure on the surface.

$$\text{energy of one photon: } E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{600 \times 10^{-9}} \approx 3.32 \times 10^{-19} \text{ J } 1.5083$$

$$\text{number of photons arriving in 1.0 s: } N = \frac{5.0 \times 10^{-3} \times 1.0}{3.32 \times 10^{-19}} \approx 1.51 \times 10^{16}$$

$$\text{momentum for one photon: } p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{600 \times 10^{-9}} \approx 1.11 \times 10^{-27} \text{ kg m s}^{-1}$$

$$\text{change of total momentum: } \Delta P = Np = 1.51 \times 10^{16} \times 1.11 \times 10^{-27} \approx 1.67 \times 10^{-11} \text{ kg m s}^{-1}$$

$$\text{average force due to these photons: } F = \frac{\Delta P}{\Delta t} = \frac{1.67 \times 10^{-11}}{1.0} \approx 1.67 \times 10^{-11} \text{ N}$$

$$\text{light pressure on surface: } p = \frac{F}{A} = \frac{1.67 \times 10^{-11}}{0.16 \times 10^{-4}} \approx 1.04 \times 10^{-6} \text{ Pa}$$

□

Question 22.8 A laser of power P is incident normally on a spot of area A , show that the pressure caused by the beam can be given by: $p = \frac{P}{cA}$.

Question 22.9 When an electron and a positron meet together, they will annihilate and produce two γ -photons: ${}^0_{-1}e + {}^0_{+1}e \longrightarrow \gamma + \gamma$. Assume the electron and the positron have negligible kinetic energy before the interaction, explain why the two photons produced must move off in opposite directions with equal energies.

22.3 wave-particle duality

22.3.1 matter waves

inspired by photon theory, which shows electromagnetic waves have a particulate nature, *Louis de Broglie* suggested in his 1924 PhD thesis that all matter has a wave-like nature ^[6]

wave characteristic of a particle can be represented by a wavelength

de Broglie wavelength of a matter particle is given by: $\lambda = \frac{h}{p}$, where $p = mv$ is the particle's momentum

Example 22.4 What is the wavelength of a human of 70 kg walking at around 2.0 m s^{-1} ?

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{75 \times 2} \approx 4.7 \times 10^{-36} \text{ m}$$

this wavelength is too small compared with any obstacle we encounter in everyday lives

so human bodies do not exhibit noticeable wave behaviour □

Example 22.5 An electron is accelerated from rest through a potential difference of 50 V. What is the de Broglie wavelength of this electron?

K.E. of electron equals change in electric P.E., so

$$\frac{1}{2}mv^2 = qV \Rightarrow v = \sqrt{\frac{2qV}{m}} = \sqrt{\frac{2 \times 1.60 \times 10^{-19} \times 50}{9.11 \times 10^{-31}}} \approx 4.19 \times 10^6 \text{ m s}^{-1}$$

$$\text{wavelength of electron: } \lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 4.19 \times 10^6} \approx 1.74 \times 10^{-10} \text{ m}$$

this wavelength is comparable to scale of atomic spacing (also around 10^{-10} m)

so these electron can be *diffracted* by solid crystals □

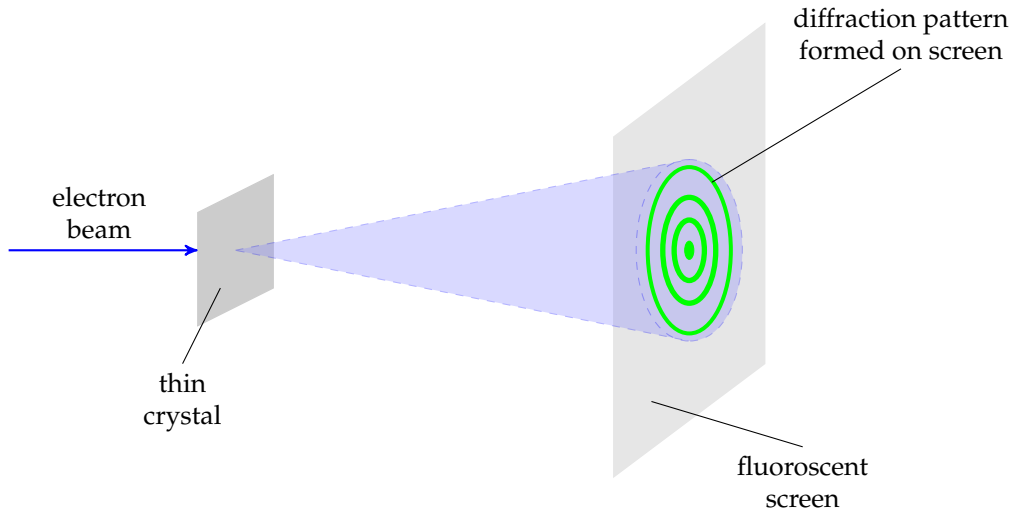
Question 22.10 An α -particle is moving with a kinetic energy of $2.4 \times 10^{-15} \text{ J}$. Find its speed, and hence find its de Broglie wavelength.

Question 22.11 If a proton and an electron are accelerated through the same voltage, (a) how would their energy compare? (b) how would their wavelength compare?

^[6] Louis de Broglie was awarded the 1929 Nobel Physics Prize 'for his discovery of the wave nature of electrons'. *Schrödinger's equation* and *Bohr's atomic model* was heavily influenced by ideas of de Broglie.

22.3.2 electron diffraction

wave property of electron was confirmed by *Clinton Davisson* and *George Thomson* in 1927
they showed experimentally electrons could be diffracted by metal crystals ^[7]



electron diffraction experiment

- electron diffraction experiment shows that particles do have wave-like properties
if electrons behaved like particles, we would see a round spot of *uniform* distribution
but the actual pattern formed on *fluorescent screen* is a set of *concentric rings*
this is a typical diffraction pattern, hence proves wave properties of electrons
- each metal has a different lattice structure, so each produces a different pattern
this allows investigation of structure of matter (explore arrangements of atoms, structures of complex molecules, structure of atomic nuclei, etc.) using electron diffraction
- since wavelength of an electron is much shorter than visible light
this allows *electron microscope* to have much higher resolving power than optical microscopes

Question 22.12 If a higher p.d. is applied to accelerate the electron beam in the electron diffraction experiment, how would the pattern change?

^[7]The Nobel Prize in Physics 1937 was awarded jointly to Clinton Davisson and George Thomson 'for their experimental discovery of the diffraction of electrons by crystals'

22.3.3 wave-particle duality

we have seen both particle-like and wave-like behaviour in light and electrons

	electromagnetic radiation	electron
particle-like behaviour	photoelectric effect, radiation pressure	deflection in electric / magnetic fields, decay, scattering
wave-like behaviour	interference, diffraction, Doppler effect, reflection, refraction	electron diffraction

when light/electron move through space, they behave like a wave

when light/electron interact with each other, they behave like particles

they show a two-sided nature, or a *dual* nature of being both a wave and a particle, described by either particle model or wave model under different circumstances

in fact, all matter (protons, neutrons, atoms, cells, basketballs, human body, earth, etc.) has this universal dual nature, called **wave-particle duality**

wave-particle duality addresses breakdown of classical concepts like particle or wave to fully describe the behaviour of microscopic objects, we need *quantum mechanics* ^[8]

22.4 quantisation of electron energy levels

in a simple atomic model, electrons move around the nucleus in circular orbits

but now we understand electrons have wave properties, as an electron moves in its orbit as a wave, it can superpose with itself

only orbits in which the electron can superpose constructively with itself are preferable

so electrons are only allowed to move in certain orbits in an atom

this means can only take certain values of energy, called *energy levels*

^[8]Being a central concept of quantum theory, wave-particle duality is deeply embedded into the foundations of quantum physics. In non-relativistic quantum mechanics, all information about a particle is encoded in its *wave function*, whose evolution with time is described by the famous *Schrödinger's equation*.

22.4.1 hydrogen atom (*)

theoretical explanation for electron energy levels was developed in 1913 by Danish physicist *Niels Bohr* in his theory of hydrogen atom ^[9]

let's look at the hydrogen atom – the simplest possible atom in nature

electrostatic attraction by proton provides cen-

tripetal force for electron to move in circles

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{m_e v^2}{r} \Rightarrow v^2 = \frac{e^2}{4\pi\epsilon_0 m_e r}$$

for electron to *constructively* superpose with itself,

the perimeter of its orbit must be an integer multiple

of its de Broglie wavelength

$$2\pi r = n\lambda \Rightarrow 2\pi r = \frac{nh}{m_e v} \Rightarrow v = \frac{nh}{2\pi m_e r}$$

where n is positive integer 1, 2, 3, ...

compare the two equations, we can eliminate v

$$\frac{e^2}{4\pi\epsilon_0 m_e r} = \frac{n^2 h^2}{4\pi^2 m_e^2 r^2}$$

we hence find radius of allowed orbits satisfies:

$$r_n = n^2 a_0 \quad \text{where } a_0 = \frac{h^2 \epsilon_0}{\pi m_e e^2} \approx 5.29 \times 10^{-11} \text{ m}$$

so we see all allowed orbits must have a radius equal to integer multiple of a_0

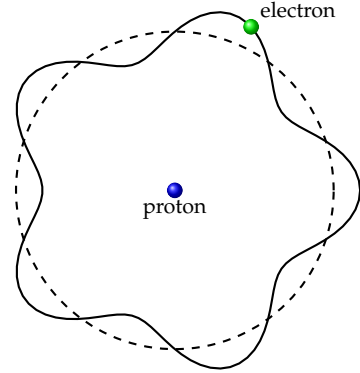
total energy possessed by the electron consist of K.E. and electric P.E.:

$$E = E_k + E_p = \frac{1}{2} m_e v^2 - \frac{e^2}{4\pi\epsilon_0 r} = \frac{1}{2} m_e \frac{e^2}{4\pi\epsilon_0 m_e r} - \frac{e^2}{4\pi\epsilon_0 r} \Rightarrow E = -\frac{e^2}{8\pi\epsilon_0 r}$$

substitute the allowed orbital radius, we find an expression for electron energy:

$$E_n = -\frac{R_E}{n^2} \quad \text{where } R_E = \frac{m_e e^4}{8h^2 \epsilon_0^2} \approx 2.18 \times 10^{-18} \text{ J} \approx 13.6 \text{ eV}$$

this shows electron can only have specific energies in the hydrogen atom



the hydrogen atom

22.4.2 electron energy levels

it can be shown that for any atom, electrons can only have certain fixed values of energy

^[9] Niels Bohr was surely one of the greatest physicists of the 20th century. He made foundational contributions to understanding atomic structure and quantum mechanics, for which he received the Nobel Physics Prize in 1922. He was also the founder of the Institute of Theoretical Physics at the University of Copenhagen, which soon became the centre of pioneering researches on quantum theory in the world.

we say energy of electrons in an atom is *discrete*, or **quantised**

these allowed values of energies are called **energy levels**^[10]

➤ *quantisation* means an electron can have only specific values of energy in an atom

no intermediate value between levels is allowed

➤ the lowest energy level is called the **ground state**
any level higher than the ground state is called an **excited state**

in our illustration, E_1 is the ground state, while E_2 , E_3 , E_4 are all excited states

➤ electron energy levels are all *negative*

to pull an electron away from the nucleus, work must be done to overcome electrostatic attraction

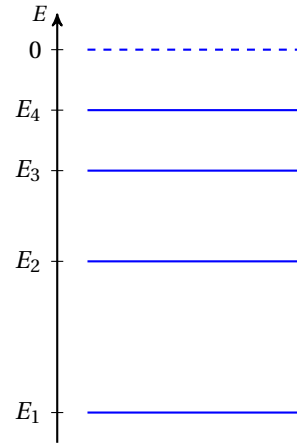
an electron at infinity would have greatest P.E., which is defined to be zero

so orbiting electrons have energies less than zero

an electron that has zero energy would become a *free* electron

➤ electrons can jump, or *transit*, between energy levels

electron transition is associated with emission or absorption of photon with the right energy



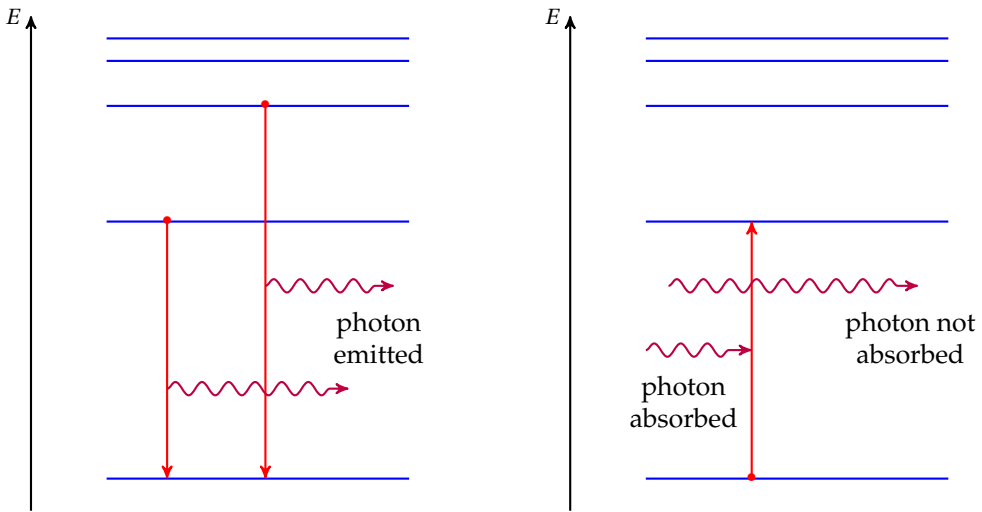
electron energy levels of some atom

22.4.3 atomic spectrum

passing light through a *prism* or a *diffraction grating*, different wavelengths are separated, a phenomenon known as **dispersion**, this produces a **spectrum** which shows distribution of energy emitted from the source in order of wavelengths

for example, white light consists of a range of wavelengths from around 400 nm to 700 nm, so it gives a *continuous spectrum* with rainbow of colours

^[10] If you study chemistry, you shall recall this idea of discrete energy levels is related to the concepts of energy shells and sub-shells of atoms.



emission and absorption of photons as a result of electron transitions



continuous spectrum of white light

electron transition between different levels causes emission or absorption of photons

this leads to the emission spectrum and absorption spectrum of atoms

emission spectrum

hot gases of an element can release photons, giving an **emission spectrum**

this happens when an electron transits from a high energy level to a lower level

energy of the photon emitted is equal to energy difference between the two levels

since electron energy levels are discrete, only specific changes of energies are possible, so only photons with specific energies can be emitted

this means photons emitted only have specific wavelengths, or specific frequencies

so a collection of sharp and bright lines are seen in emission spectrum

- emission spectrum is a *discrete* spectrum, also called a *line* spectrum
- atoms of each element has a unique electron energy level structure



emission spectrum of the hydrogen atom

so each element produces a unique emission spectrum, leading to quite a few applications:

- explains the colour of flames when a particular chemical element is present
- allows the identification of elements in an unknown substance in chemical analysis
- explains varied colours of electric signs lighted by gas-discharge tubes

absorption spectrum

pass white light through a *cool gas*, photons with the right energies can be absorbed, giving rise to an **absorption line spectrum**

by absorbing photon, electron can transit from a low energy level to a higher level

energy of the photon absorbed must equal energy difference between the two levels

since electron energy levels are discrete, so only photons with specific energies can be absorbed, while other photons are unaffected as they pass through the gas

wavelengths of these absorbed photons will be missing in the emergent spectrum

so we would observe a set of *dark lines* appearing in a background of continuous spectrum

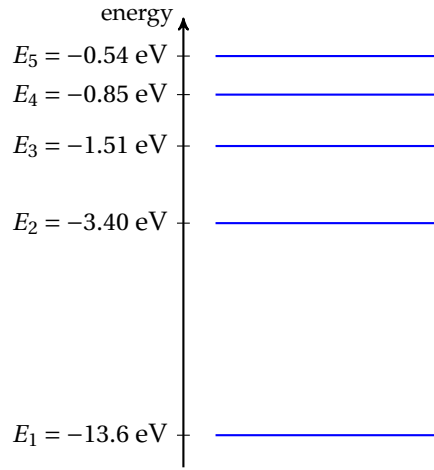


absorption spectrum for some element

- absorption spectrum is also a *discrete* spectrum, or a *line* spectrum
- as photons with the right energies are absorbed, they can be re-emitted through *de-excitation* but these photons are re-emitted in *random* directions so they would still appear dark compared with those unaffected wavelengths
- absorption spectrum is widely used in many areas
 - determine the composition of a particular substance in analytical chemistry

- determine chemical compositions of stars in astronomical spectroscopy
- explain colour of chemicals in terms of the complementary colour of photons absorbed

Example 22.6 Some electron energy levels of the hydrogen atom is shown. Find the longest and the shortest wavelength produced by electron transitions between the energy levels given.



energy levels of the hydrogen atom

energy of photon emitted equals change of electron energy level: $\frac{hc}{\lambda} = \Delta E$, or $\lambda = \frac{hc}{\Delta E}$

transition with least/greatest energy change gives rise to longest/shortest wavelength, so

$$E_5 \rightarrow E_4 \Rightarrow \lambda_{\max} = \frac{hc}{E_5 - E_4} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{(0.85 - 0.54) \times 1.60 \times 10^{-19}} \approx 4.0 \times 10^{-6} \text{ m} \quad (\text{infra-red})$$

$$E_5 \rightarrow E_1 \Rightarrow \lambda_{\min} = \frac{hc}{E_5 - E_1} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{(13.6 - 0.54) \times 1.60 \times 10^{-19}} \approx 9.5 \times 10^{-8} \text{ m} \quad (\text{ultraviolet}) \quad \square$$

Example 22.7 The emission spectrum of the hydrogen atom consists of a number of wavelengths in the visible spectrum. Given that the visible spectrum consists of light of wavelengths within the range from 380 nm to 740 nm, also use data in Example 22.6, find the energies of visible photons that can be produced by transitions between the energy levels shown.

let's first find the range of energies, in eV, for visible photons

$$E_{\max} = \frac{hc}{\lambda_{\min}} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{380 \times 10^{-9}} \approx 5.23 \times 10^{-19} \text{ J} = \frac{5.23 \times 10^{-19}}{1.60 \times 10^{-19}} \text{ eV} \approx 3.27 \text{ eV}$$

$$E_{\min} = \frac{hc}{\lambda_{\max}} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{740 \times 10^{-9}} \approx 2.69 \times 10^{-19} \text{ J} = \frac{2.69 \times 10^{-19}}{1.60 \times 10^{-19}} \text{ eV} \approx 1.68 \text{ eV}$$

we shall find the combinations of electron energy levels such that their difference fall within the range between 1.68 eV and 3.27 eV, by trial and error, we find three possible combinations:

$$(1) E_3 \rightarrow E_2 \Rightarrow 3.40 - 1.51 = 1.89 \text{ eV}$$

$$(2) E_4 \rightarrow E_2 \Rightarrow 3.40 - 0.85 = 2.55 \text{ eV}$$

$$(3) E_5 \rightarrow E_2 \Rightarrow 3.40 - 0.85 = 2.85 \text{ eV}$$

these emission lines are called H_α (656 nm), H_β (486 nm) and H_γ (435 nm) lines^[11]

they are three of the four the hydrogen emission lines that are visible to human eyes^[12] $\square$

Example 22.8 A white light is incident on a cloud of cool hydrogen gas. In the emergent spectrum, a dark line is observed at a wavelength of 435 nm. Determine the energy change that gives rise to this dark line.

$$\text{photon absorbed: } E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{435 \times 10^{-9}} \approx 4.57 \times 10^{-19} \text{ J} = \frac{4.57 \times 10^{-19}}{1.60 \times 10^{-19}} \text{ eV} \approx 2.86 \text{ eV}$$

note that $3.40 - 0.54 = 2.86 \text{ eV}$, so this dark line is due to the electron transition: $E_2 \rightarrow E_5$ $\square$

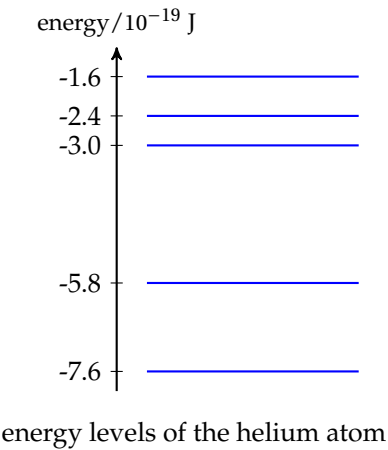
Question 22.13 If we only consider the electron energy levels given in Example 22.6, how many different wavelengths can be detected in the emission spectrum of the hydrogen atom ?

Question 22.14 Three lines are observed at wavelengths 486 nm , 656 nm, and 1880 nm in the emission spectrum of hydrogen atoms. (a) Calculate the photon energies for these wavelengths. (b) Draw a diagram with *three* labelled energy levels, and show the energy changes for the three wavelengths produced with arrows in your diagram.

^[11]The H_α line is the brightest hydrogen line in the visible range. It plays an important role in astronomy, as it can be used to study a star's surface temperature, the velocity of a distant stellar object, etc.

^[12]These lines belong to a family of the spectral lines of the hydrogen atom, known as the *Balmer lines*, named after Johann Balmer. Balmer discovered an empirical equation to predict the series in 1885, but the reason why the equation worked was eventually clarified by Neils Bohr with the atomic model which now bears his name. We now understand the Balmer lines correspond to emissions of photons by electrons jumping to the second level from higher energy states.

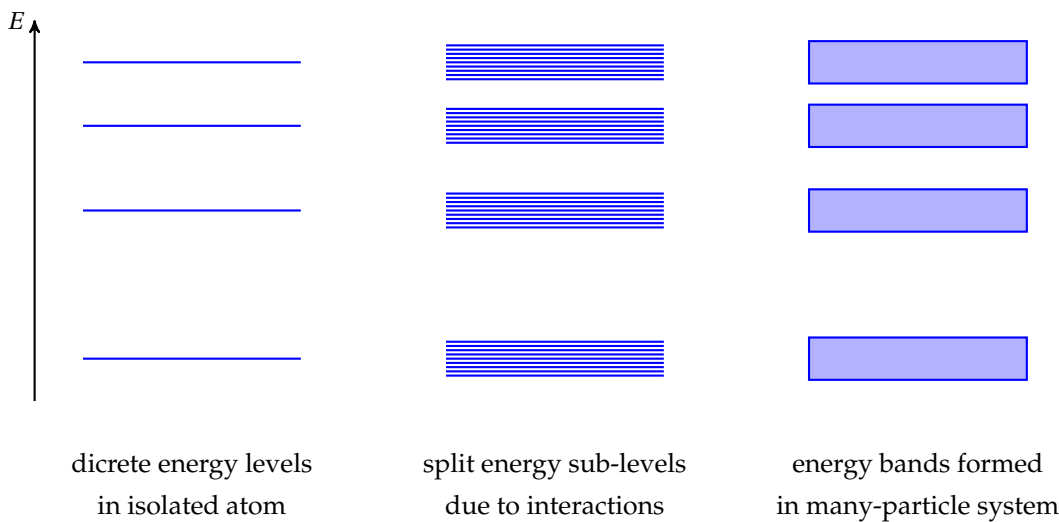
Question 22.15 The relative cool atmosphere of the sun could give rise to dark lines in the spectrum of sunlight. One particular dark spectral line has a wavelength of 590 nm. By reference to the energy levels of the helium atom, suggest how this dark line provides evidence of the presence of helium in sun’s atmosphere. You may draw an arrow to show the possible electron transition that gives rise to this dark line.



22.5 band theory (*)

22.5.1 energy bands

- in an isolated atom, electrons have discrete energy levels
- in solids, interaction between neighbouring atoms causes change of energies
- original energy levels split into a band with many sub-levels
- number of sub-levels in each band equals number of atoms in solid in general
- due to very large number of atoms, sub-levels (seem to) form *continuous* **energy bands**



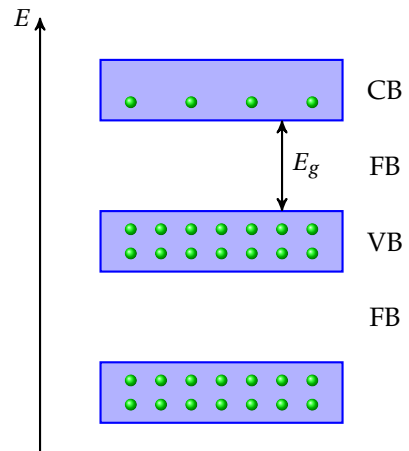
in solids, electrons fill up from lowest energy bands available (see figure for a rough idea)

forbidden bands (FB) represent energies that electrons are not allowed to take

highest fully filled band is called **valence band** (VB)

next partially filled or empty band is called **conduction band** (CB)

separation between valence band and conduction band defines **band gap** (E_g) of material



22.5.2 band theory & electrical conductivity

electrical conductivity of material can be explained by its band structure

low-energy bands below VB are usually fully filled at all times, all states are occupied, electrons cannot move freely to form electric currents, so we are not interested in low-energy bands

conductivity of material will depend on whether there are free *charge carriers* in CB and VB

metallic conductors

for typical metals, CB *overlaps* with VB, i.e., there is no band gap

there exist vacant states for conduction electrons to occupy at ease

this means conduction electrons are free to move around to form currents

as temperature rises, there is no significant change in number of free electrons

but *lattice vibration* of atoms increases, electrons become more likely to collide with vibrating atoms, so resistance of metal increases with temperature

➤ metals are *opaque* to visible light or other low-frequency radiation

width of CB for a typical metal is about 2 ~ 3 eV

low-energy photons can be absorbed by conduction electrons

visible light, infra-red, microwave therefore cannot pass through metal

➤ metals are *transparent* to high-frequency radiation such as X-rays

if X-ray photons were absorbed, electrons would take energies in forbidden band but this is not allowed, so high-frequency photons penetrate through metal

insulators

examples of insulators include glass, diamond, etc.

in normal conditions, CB of insulator is empty, energy bands up to VB are fully filled, so no movement of electron is possible, material has poor conductivity

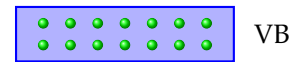
insulators also have huge band gap^{[13][14]}, so thermal excitations cannot easily make VB electrons jump into CB

insulators remain poor conductors as temperature rises

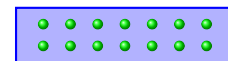
Question 22.16 Diamond has a band gap $E_g \approx 6.0$ eV. By reference to its electronic band structure, explain why diamond appears *transparent* to visible light?



CB



VB



band structure of insulator

intrinsic semiconductors

examples of an intrinsic semiconductor are silicon and germanium

band structure of an intrinsic semiconductor is similar to that of insulators

at low temperature, no free electron from the empty CB and completely occupied VB, so semi-conductors are not conductive in normal conditions

however, semiconductors have much narrower band gaps

as temperature rises, VB electrons gain thermal energy to cross band gap and enter CB

the electrons jumped into CB surely can move around to form electric currents

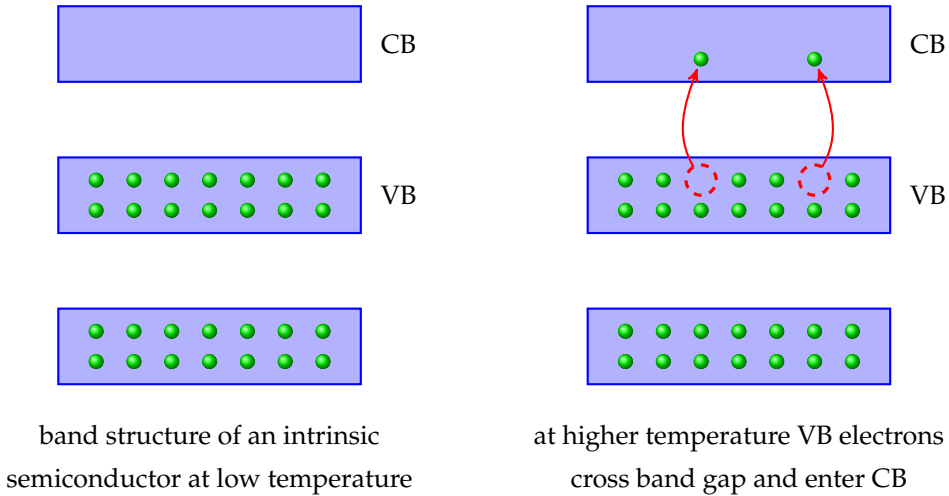
at the same time, VB is no longer completely filled, *holes* are formed

^[13]To be a little more precise, materials with a band gap $E_g \approx 5$ eV or greater are regarded as insulators.

^[14]There exist a class of insulators called *Mott insulators* which do not have large band gaps. Their poor conductivity at low temperatures is due to electron-electron interactions, which are not considered under conventional band theories.

a hole is a site where an electron is missing, neighbouring electrons can move to fill the hole and leave a new hole, but this can be thought of as the hole moving about

there are now more *charge carriers* (negatively-charged electrons and positively-charged holes) available, so semiconductor has better conductivity, resistance of material would decrease



➤ *lattice vibration* could also affect resistance of semiconductors

as temperature rises, vibration of atoms increases, charge carrier are more likely to get scattered, causing a decrease in conductivity

but the effect of more charge carriers is far greater than effect due to lattice vibration

so resistance of semiconductor decreases at higher temperature

Example 22.9 Silicon has a band gap $E_g \approx 1.1$ eV. State and explain whether it becomes conducting when exposed to red light of 600 nm.

🔗 energy of radiation: $E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{600 \times 10^{-9}} \approx 3.32 \times 10^{-19} \text{ J} \approx 2.1 \text{ eV} > E_g$

this energy is sufficient to make valence electrons in silicon to cross band gap

silicon now possesses conducting electrons and holes therefore becomes conducting □

Question 22.17 Use band theory to explain why an LDR (light-dependent resistor) has a very high resistance in the dark, but its resistance drops dramatically when being exposed to light.